



CSE422 Project Report

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1. Introduction & Problem Definition

In the digital age, understanding user emotions through text is critical for applications in social media analysis, mental health monitoring, customer service, and more. However, for low-resource languages like Bangla and Banglish which integrates Bengali syntax with Roman script, there has been a significant gap in emotion detection resources. This project aims to evaluate multiple transformer based pre-trained models on a large-scale emotion-labeled dataset for both Bangla and Banglish.

We focus on classifying text into six basic emotions defined by Ekman: joy, sadness, anger, fear, disgust, and surprise. The core challenge lies in handling Banglish, class imbalance and ensuring emotional accuracy across both formal (Bengali) and informal (Banglish) texts.

2.Dataset Description

The dataset used in this project is a large-scale resource designed for multi-class emotion classification in Bengali and Banglish (Bengali transliterated into Latin script). Each text sample is labeled with one of Ekman's six basic emotions—anger, disgust, fear, joy, sadness, and surprise—making the task a single-label, six-class classification problem.

The complete dataset consists of 80,098 text samples, evenly split into Bengali and Banglish versions. While some portions were sourced from pre-existing datasets with labels, a significant portion—38,670 cleaned web-scraped entries from YouTube and Twitter focusing on popular domains among Bangladeshi people: politics, sports, business, education, technology, health, religion, and entertainment and they were manually annotated by native Bengali speakers. These annotations were validated using a majority voting technique and expert review, resulting

in a high Cohen's Kappa score of 93.5%, indicating strong inter-annotator agreement and label consistency.

To create the Banglish version, all Bengali samples were manually transliterated by the same annotators and cross-validated for spelling consistency. This dual-format structure enhances the dataset's utility in low-resource language research and multilingual NLP.

Despite its high quality, the dataset exhibits a slight class imbalance, with certain emotions with joy accounting for the highest instances at 17,836 and fear for the lowest at 7,565

The distribution of emotion classes is as follows:

Emotion	Number of Samples (Per Language)
Joy	17836
Sadness	16309
Anger	15179
Disgust	13098
Surprise	10107
Fear	7565

Overall, the dataset offers a rich resource for training robust emotion detection models and holds great promise for practical applications such as sentiment monitoring, mental health tracking, customer service, and content moderation, particularly in the context of low-resource South Asian languages.

3. Dataset Preprocessing

To prepare the Bengali and Banglish emotion classification dataset for transformer-based models, we developed a CustomDataset class using PyTorch. This class efficiently handles text preprocessing and tokenization, ensuring the data is compatible with models such as BanglaBERT, XLM-RoBERTa etc.

3.1 Text Cleaning

Each input text is first cleaned by removing any unnecessary or redundant whitespace. This step standardizes the format of the textual data, eliminating inconsistencies that may hinder tokenization or model interpretation. Empty or duplicate rows were removed.

3.2 Label Encoding

Emotion labels (originally strings like "joy", "anger", etc.) were converted into numeric labels using LabelEncoder.

3.3 Tokenization

A custom PyTorch Dataset class named **CustomDataset** was developed to efficiently serve batches of tokenized data during training and evaluation. For the **BanglaT5** model, the **normalize_bangla_text** function was applied exclusively to the Bengali dataset to align the input with the clean and standardized text the model was trained on, thereby enhancing its understanding and overall performance. This dataset design integrates seamlessly with PyTorch's DataLoader, which facilitates the creation of train, validation, and test loaders with support for batching and shuffling during training.

We employed the appropriate tokenizer from Hugging Face's Transformers library depending on the model (e.g., BanglaBERT or XLM-RoBERTa). The tokenizer is responsible for converting raw text into model-ready numerical representations. The key tasks performed during tokenization include:

- **Adding special tokens** (e.g., [CLS], [SEP]) required by BERT-based architectures.
- **Padding and truncating** sequences to a fixed length (max_len=128) to ensure uniform input sizes.
- **Generating attention masks** to help the model distinguish between actual tokens and padded content.
- **Returning token type IDs**, especially relevant for models trained on sentence-pair tasks.

This is done through the `encode_plus()` function, which returns the necessary inputs in tensor format.

3.4 Output Format

The `__getitem__` method of the class returns a dictionary with the following elements for each text instance:

- `input_ids`: The tokenized and encoded representation of the input text.
- `attention_mask`: A binary mask identifying non-padded versus padded tokens.
- `token_type_ids`: Segment embeddings, useful for certain transformer models.
- `targets`: The true class label (as an integer), representing one of the six emotion categories.
- `text`: The original cleaned version of the input text, useful for error analysis or visualization.

This preprocessing pipeline ensures that every text entry in the dataset is transformed into a consistent, high-quality input format suitable for training deep learning models. It allows us to

fully leverage the capabilities of modern transformer architectures in emotion classification tasks involving Bengali and Banglish language data.

3.5 Dataset Splitting

The dataset was split into training (80%), validation (10%), and test (10%) sets, with stratified sampling to preserve class distribution.

4. Model Descriptions

4.1 BanglaBERT

Overview:

BanglaBERT is a monolingual transformer-based language model specifically built for the Bangla language. It follows the BERT-base architecture and addresses the scarcity of high-quality NLP tools for Bangla by offering a robust, language-specific pretrained model. It has become a foundational resource in Bangla NLP tasks such as sentiment analysis, hate speech detection, and emotion classification.

Architecture:

- Model Type: BERT-base, Type: Encoder-only
- Transformer Layers: 12
- Hidden Size: 768
- Attention Heads: 12
- Parameters: ~110 million
- Tokenizer: SentencePiece tokenizer optimized using the BNLP package (vocabulary size ~102,025 tokens)
- Pretraining Objective: Masked Language Modeling (MLM)
- Training Corpus: 18.6 GB of native Bangla data, including Wikipedia, news articles, and blogs.

Advanced Highlights:

1. Tokenization Uniqueness with SentencePiece: Unlike mBERT or multilingual models that tokenize based on whitespace and punctuation, BanglaBERT uses a SentencePiece tokenizer. This tokenizer is better suited to the morphological structure of Bangla, allowing effective tokenization of compound words and inflected forms. Bangla is morphologically rich, with features like gendered suffixes and verb inflections. BanglaBERT's tokenizer helps preserve semantic structure, improving understanding in downstream tasks.

2. Code-Mixed Text Generalization: Despite not being trained directly on Romanized Bangla (Banglish), BanglaBERT demonstrates solid performance on code-mixed text after fine-tuning. This is attributed to the presence of English loanwords and phrases in the Bangla Wikipedia corpus used during pretraining.: Real-world data (e.g., social media, chat) often includes mixed-language content. A monolingual model's adaptability to such data is rare and valuable.

3. Smaller Size, Competitive Results: With ~110M parameters, BanglaBERT is significantly smaller than multilingual models like XLM-R (270M) or mBERT, yet achieves superior results on Bangla tasks. Smaller, domain-specific models can outperform larger multilingual ones when fine-tuned properly — especially in resource-limited settings.

4. Community-Centric Development: BanglaBERT was developed independently by Sagor Sarker without institutional backing, making it the first open-access large-scale Bangla model. It has catalyzed local NLP research and is widely used in academic and applied projects across Bangladesh.

Limitations:

- Built solely for Bangla — underperforms in multilingual or cross-lingual contexts.
- Not ideal for handling Romanized Bangla or other non-native scripts without fine-tuning.
- Smaller training corpus compared to XLM-R, limiting its generalization beyond Bangla.
- May reflect topical and stylistic biases due to limited domain diversity in the training data.

Use Cases:

- Emotion and sentiment analysis (e.g., tweets, reviews)
- Hate speech classification
- Named Entity Recognition (NER)
- Bangla question answering and classification tasks
- Educational and low-resource mobile NLP applications

4.2 XLM-RoBERTa

Overview:

XLM-RoBERTa is a large-scale multilingual transformer model developed by Facebook AI, designed to handle 100 languages with a unified architecture. It is an extension of RoBERTa, optimized for cross-lingual understanding and trained on massive multilingual corpora. Its architecture and training objective make it highly effective for zero-shot and multilingual NLP applications, including those involving low-resource languages like Bangla.

Architecture:

- Model Type: RoBERTa-base (also available in a larger variant), Encoder-only
- Transformer Layers: 12
- Hidden Size: 768
- Attention Heads: 12
- Parameters: ~270 million
- Tokenizer: SentencePiece (Byte-Pair Encoding, language-agnostic)
- Pretraining Objective: Masked Language Modeling (MLM)
- Training Corpus: 2.5 TB of filtered CommonCrawl data across 100 languages

Key Features:

- Trained on a much larger and more diverse dataset compared to mBERT (over 10× more data)
- No language-specific embeddings — the model learns language-agnostic patterns, improving generalization
- Strong performance on cross-lingual tasks including zero-shot learning and transfer tasks
- Effectively handles code-mixed and Romanized text like Banglish

Performance & Strengths:

- Outperforms mBERT in key benchmarks like XNLI, MLQA, and Universal NER
- Shows significant gains for low-resource languages due to better representation in training corpus
- Enables cross-lingual sentiment analysis, multilingual QA, and entity recognition without language-specific retraining
- Capable of zero-shot transfer between high-resource and low-resource languages — fine-tune on English and apply to Bangla or vice versa

Advanced Highlights:

- Cleaned and Filtered Pretraining Data: Its dataset is curated to remove duplicates and noise, providing richer linguistic diversity

- No Language Embeddings: Improves performance in multilingual and code-switched contexts by eliminating language-specific cues
- Zero-Shot Learning Capabilities: Allows robust performance across unseen languages without retraining on the target language
- Efficient Base Model: The base version (270M parameters) achieves strong results, often sufficient for classification tasks, while being less resource-intensive than the large version

Limitations:

- Not specifically fine-tuned for Bangla — may underperform compared to BanglaBERT in Bangla-only emotion or classification tasks
- Lacks deep morphological sensitivity that a native Bangla model like BanglaBERT can provide
- Some bias towards high-resource languages persists in early model layers, requiring careful fine-tuning for equitable Bangla performance.

Use Cases:

- Cross-lingual sentiment analysis
- Multilingual question answering
- Language detection
- Named entity recognition across languages
- Code-switched and Romanized text classification (e.g., Banglish)

4.3 BanglaT5

BanglaT5 is a sequence-to-sequence (encoder-decoder) Transformer model, inspired by Google's T5 architecture. It is pre-trained using a span corruption objective, where spans of text are masked and the model learns to predict them. This approach enables the model to handle various natural language generation (NLG) tasks effectively.

The model is trained exclusively on Bengali text. Specifically, it was pre-trained on a 27.5 GB corpus of Bangla data, collected from a meticulously selected list of web sources.

Expertise

BanglaT5 excels in Bangla NLG tasks, achieving state-of-the-art results in:

- Machine Translation.
- Abstractive Text Summarization.
- Question Answering.
- Multi-Turn Dialogue Generation.

- News Headline Generation.
- Cross-lingual Summarization.

In evaluations, BanglaT5 outperformed several multilingual models, including mT5 (base), XLM-ProphetNet, mBART-50, and IndicBART, particularly in tasks like machine translation and question answering.

Limitations

- Being monolingual, BanglaT5 is not suitable for tasks involving other languages.
- It requires a specific text normalization pipeline before tokenization to achieve optimal performance.

4.4 mT5

mT5 is a multilingual variant of the T5 model, employing the same encoder-decoder Transformer architecture. It is trained using a masked span prediction objective, similar to T5, allowing it to handle a wide range of text-to-text tasks across multiple languages.

mT5 is trained on the mC4 corpus, a multilingual version of the C4 dataset, encompassing 101 languages. This extensive training enables mT5 to perform tasks in various languages, including low-resource ones.

Expertise

mT5 demonstrates strong performance in:

- Multilingual Machine Translation
- Question Answering
- Text Summarization
- Natural Language Inference.

Its ability to handle multiple languages makes it versatile for cross-lingual tasks and applications involving code-switching.

Limitations

mT5 may underperform in tasks requiring fine-grained understanding of specific languages, especially those underrepresented in its training data.

For tasks like Named Entity Recognition (NER), mT5's performance is generally weaker compared to models like XLM-R, unless larger versions (e.g., mT5-XL) are used.

5.Results

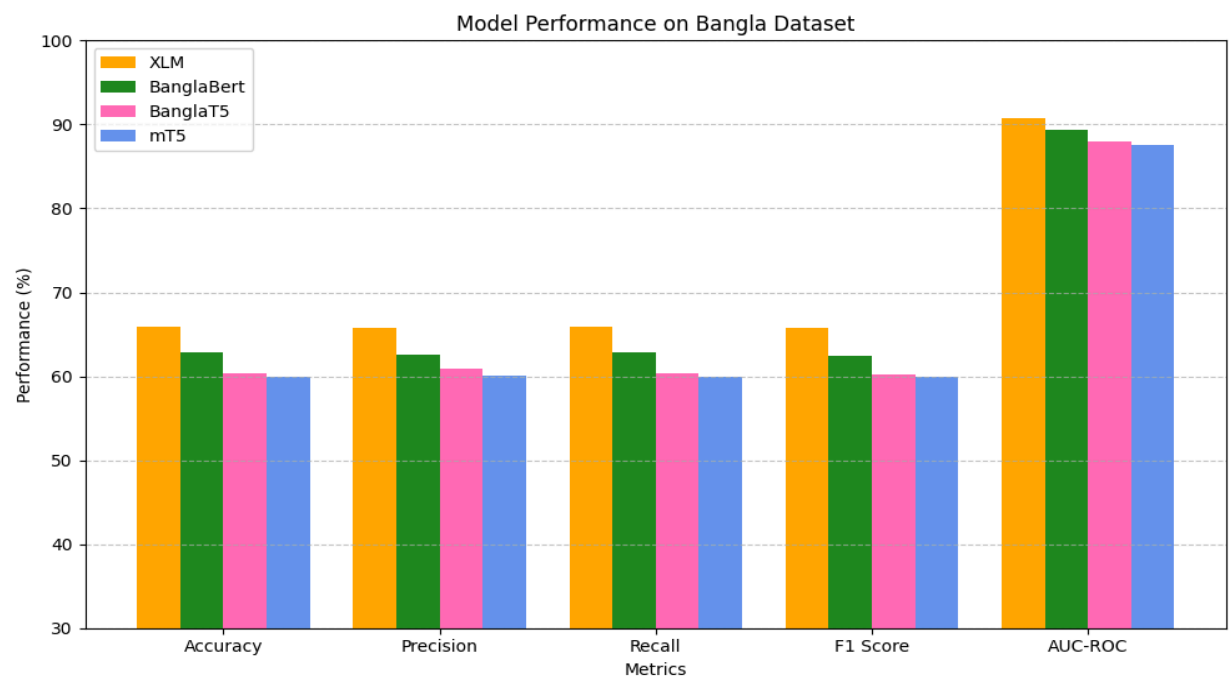


Figure 1

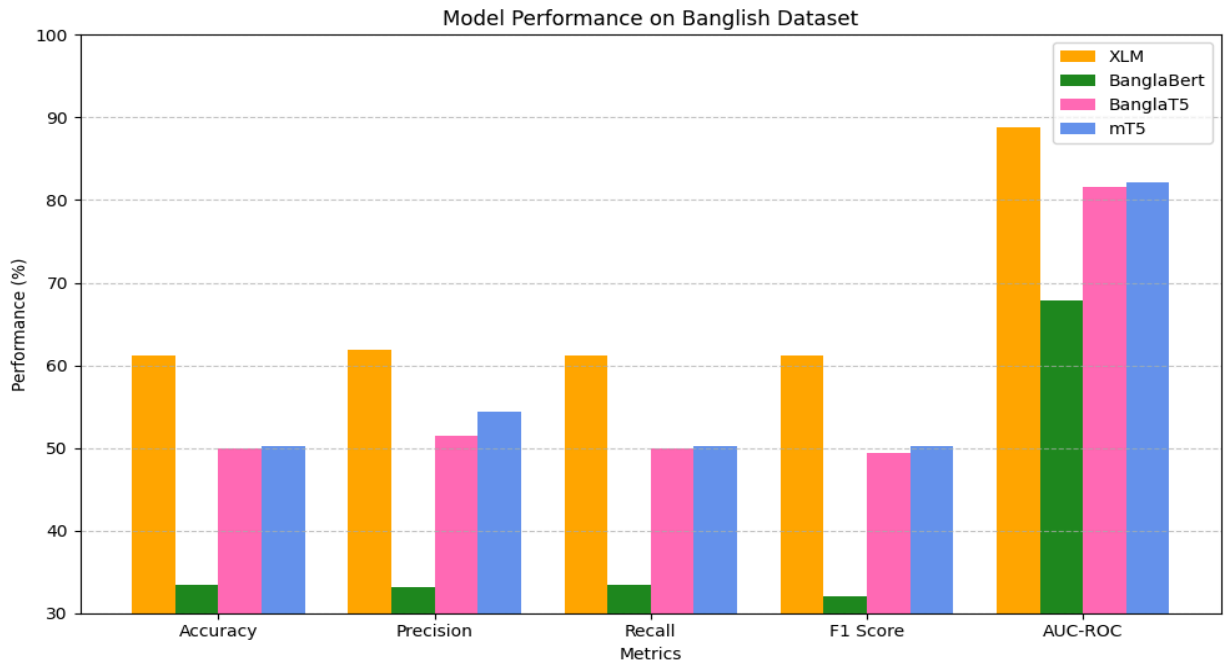


Figure 2

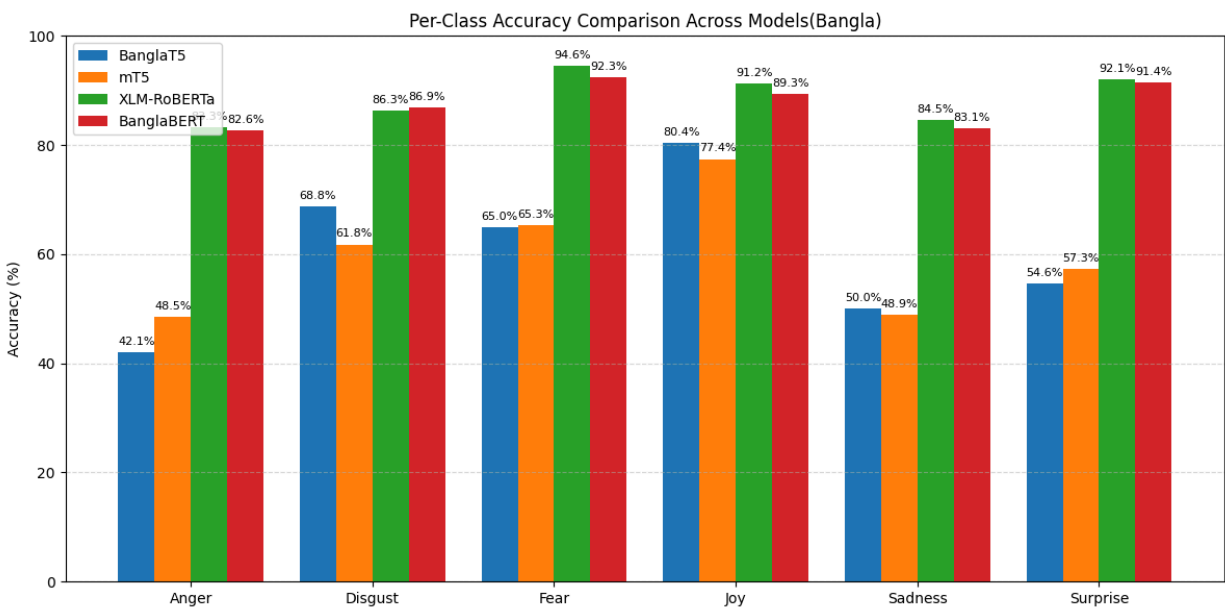


Figure 3

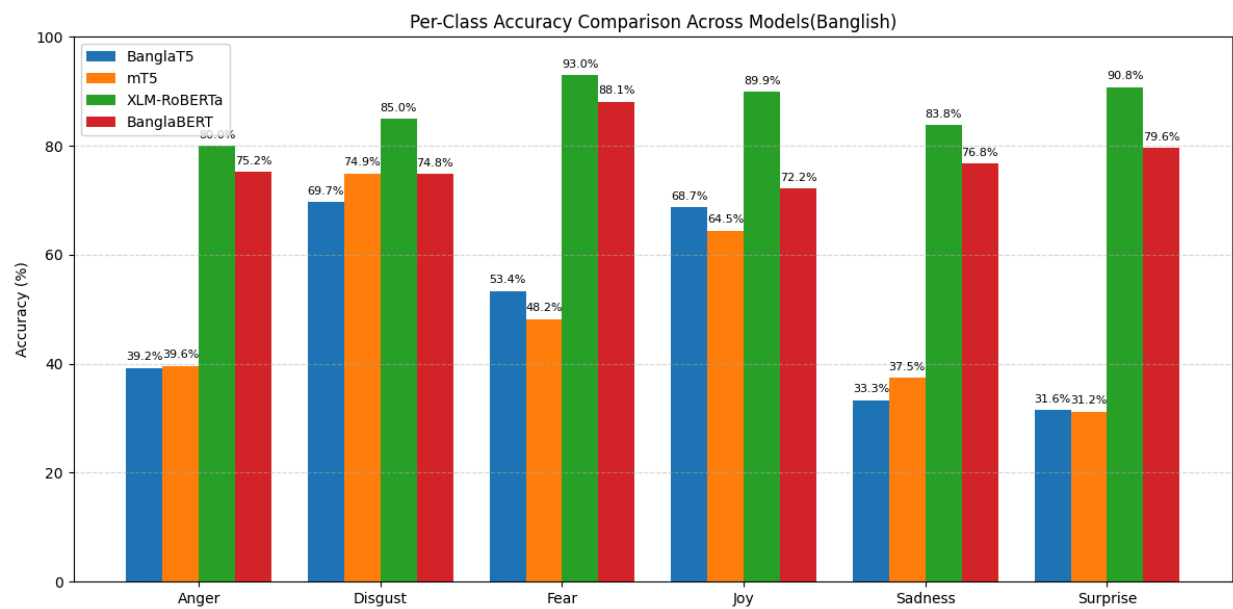


Figure 4

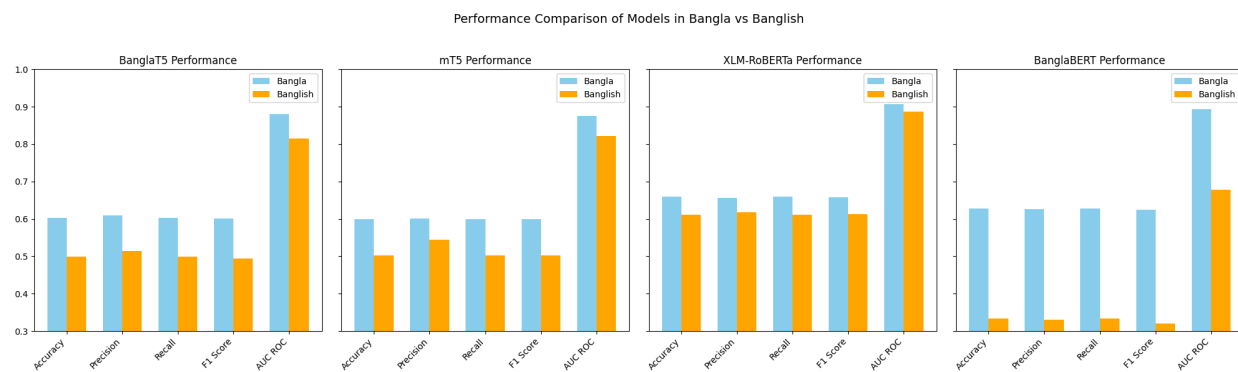


Figure 5

Analysis

Here's a clear side-by-side comparison table for all models across all evaluation metrics for both **Banglish** and **Bangla** datasets:

Model	Dataset	Accuracy	Precision	Recall	F1 Score	AUC-ROC
XLM	Banglish	61.19	61.83	61.19	61.21	88.77
BanglaBERT	Banglish	33.41	33.11	33.41	32.02	67.82
BanglaT5	Banglish	49.93	51.42	49.93	49.35	81.55
mT5	Banglish	50.26	54.37	50.26	50.17	82.14
XLM	Bangla	65.94	65.70	65.94	65.75	90.76
BanglaBERT	Bangla	62.82	62.59	62.82	62.45	89.35
BanglaT5	Bangla	60.35	60.94	60.35	60.17	87.98
mT5	Bangla	59.89	60.13	59.89	59.87	87.55

Here is a detailed explanation of the graph titled "**Model Performance on Bangla Dataset**" in Figure 1:

Across all evaluation metrics, XLM-RoBERTa consistently outperforms the other models, demonstrating the most robust and reliable performance. Its strengths are especially notable in Accuracy, Precision, Recall, and F1 Score, where it leads by a margin of 2–3% over the next best model. In the AUC-ROC metric, all models perform well, with XLM-RoBERTa slightly ahead.

- XLM-RoBERTa achieves the highest performance across all metrics, including an exceptional AUC-ROC score above 91%.
- BanglaBERT is a strong second, with consistent results across metrics and an AUC-ROC close to 90%.
- BanglaT5 and mT5 perform similarly, with slightly lower scores (~60–61%) for Accuracy, Precision, Recall, and F1 Score, but still respectable AUC-ROC values (~88–89%).

XLM-RoBERTa dominates across all metrics, indicating robust performance on native Bangla text. BanglaBERT, while trailing slightly behind XLM, still performs competitively—especially in AUC-ROC and Accuracy—showing its alignment with Bangla linguistic features. BanglaT5 and mT5, although designed with multilingual capabilities, perform slightly lower across all metrics, suggesting that they may not fully leverage Bangla-specific linguistic cues in comparison to XLM-RoBERTa and BanglaBERT.

The AUC-ROC scores are significantly higher for all models, indicating excellent classification capability when measured across threshold ranges—even when other metrics (like Precision/Recall) are moderate.

This graph validates that XLM-RoBERTa is the most effective model for Bangla emotion classification, with consistent top scores across Accuracy, Precision, Recall, F1 Score, and AUC-ROC. BanglaBERT also shows strong performance, particularly in tasks closer to its pretraining objective. The close scores of BanglaT5 and mT5 suggest that while they are adequate, they may not be as finely tuned for the Bangla dataset as the other two models.

The provided graph in Figure 2 titled "Model Performance on Banglish Dataset" presents a comprehensive comparison of four models—XLM-RoBERTa (XLM), BanglaBERT, BanglaT5, and mT5—across five key performance metrics: Accuracy, Precision, Recall, F1 Score, and AUC-ROC.

- XLM-RoBERTa outperforms all other models across every metric, achieving the highest scores in Accuracy (~62%), Precision, Recall, and F1 Score (~61%), as well as an outstanding AUC-ROC (~89%).
- mT5 and BanglaT5 show comparable mid-range performance, scoring around 50–55% across core metrics, and performing decently in AUC-ROC (~81–82%).
- BanglaBERT, while showing some improvement in AUC-ROC (~67%), performs the weakest across all other metrics (~33–34%), suggesting limited effectiveness in handling Banglish (code-mixed) data

XLM-RoBERTa clearly dominates in all metrics on the Banglish dataset, particularly excelling in AUC-ROC, which highlights its strong generalization across classification thresholds.

mT5 and BanglaT5 deliver moderate and fairly balanced performances across metrics, making them reliable, though not outstanding. BanglaBERT, while potentially optimized for native Bangla, performs significantly worse in this Banglish context—especially in Accuracy, Precision, and Recall—indicating possible limitations in handling code-mixed or Romanized input.

This graph supports the conclusion that XLM-RoBERTa is the most effective model for Banglish emotion classification, while BanglaBERT's design might not align well with Banglish linguistic

patterns. The AUC-ROC dominance of XLM-RoBERTa also suggests its suitability for real-world applications where classification thresholds may vary.

Per-Class Accuracy Comparison Analysis (Bangla & Banglish)

Bangla Dataset:

The first graph shows the per-class accuracy of four models — BanglaT5, mT5, XLM-RoBERTa, and BanglaBERT — across six emotion classes: *Anger*, *Disgust*, *Fear*, *Joy*, *Sadness*, and *Surprise*.

- XLM-RoBERTa consistently outperforms the other models across all classes, achieving the highest accuracy except the Disgust class.
- BanglaBERT closely follows, particularly in *Fear* (92.3%) and *Joy* (89.3%), showing competitive performance but for disgust, it outperforms the other models,
- BanglaT5 and mT5 lag behind, especially in *Anger* (42.1% and 48.5% respectively), though they remain more competitive in *Fear* and *Sadness*.

Banglish Dataset:

The second graph illustrates performance on the Banglish dataset, which presents greater linguistic complexity due to code-mixed input.

- XLM-RoBERTa again leads in performance, particularly excelling in all the classes for example *Fear* (93.0%), *Joy* (89.9%), and *Surprise* (90.8%).
- BanglaBERT performs robustly but slightly behind XLM-RoBERTa in most classes, showing resilience even in the more complex Banglish setting.
- BanglaT5 and mT5 show noticeably reduced performance compared to their Bangla dataset results, especially in classes like *Sadness* and *Surprise* (around 30–35%).

XLM-RoBERTa is the most robust model across both Bangla and Banglish datasets, particularly strong in detecting high-intensity emotions like *Fear*, *Joy*, and *Surprise*.

BanglaBERT also performs consistently well, though it is slightly less accurate than XLM-RoBERTa.

Transformer-based generative models (BanglaT5 and mT5) underperform in code-mixed (Banglish) settings, highlighting a need for better adaptation to noisy or mixed-language data.

The image (Figure 5) shows four bar charts comparing performance metrics for various models using Bangla (blue bars) and Banglish (orange bars) languages. Each graph displays results across five evaluation categories: Accuracy, Precision, Recall, F1-score, and AUC-ROC.

BanglaT5 Performance (Leftmost Graph)

In the BanglaT5 model:

- Bangla consistently outperforms Banglish across all metrics
- The largest performance gap appears in Accuracy (~0.6 vs ~0.5)
- Both languages perform best in the AUC-ROC metric (~0.85 for Bangla, ~0.8 for Banglish)
- For Precision, Recall, and F1-score, Bangla maintains around 0.6 while Banglish averages around 0.5
- The performance difference is relatively consistent across all evaluation metrics except AUC-ROC, where both languages show significantly improved performance

mT5 Performance

For the mT5 model:

- Similar pattern to BanglaT5, with Bangla outperforming Banglish
- Performance gaps are slightly smaller than in BanglaT5
- Again, both languages show the highest scores in AUC-ROC (~0.85 for Bangla, ~0.8 for Banglish)
- Banglish shows slightly better performance in Precision compared to other metrics
- Both languages show consistent performance across Accuracy, Precision, Recall, and F1-score (~0.6 for Bangla, ~0.5 for Banglish)

XLM-RoBERTa Performance

The XLM-RoBERTa model shows:

- Smaller performance gaps between Bangla and Banglish
- Overall higher performance for Banglish compared to the other models
- Both languages perform extraordinarily well on AUC-ROC (~0.9 for both)
- Across Accuracy, Precision, Recall, and F1-score, both languages maintain consistent performance of ~0.65 for Bangla and ~0.6 for Banglish
- This model shows the most balanced performance between the two languages

BanglaBERT Performance

For the BanglaBERT model:

- Shows the largest performance gap between Bangla and Banglish
- Bangla performance is consistently around 0.6 across Accuracy, Precision, Recall, and F1-score
- Banglish performs notably worse on all metrics except AUC-ROC, with scores around 0.3-0.35
- Both languages still perform best on AUC-ROC (~0.85 for Bangla, ~0.68 for Banglish)
- This model appears most specialized for native Bangla language processing

Overall Observations

- Across all models, Bangla consistently outperforms Banglish
- The AUC-ROC metric shows the highest performance for both languages in all models
- XLM-RoBERTa shows the smallest performance gap between languages
- BanglaBERT shows the largest performance gap, particularly favoring Bangla
- Performance patterns remain relatively consistent across metrics within each model
- The transformer-based models (particularly XLM-RoBERTa) seem to handle both languages more effectively than others

This suggests that while native Bangla language processing is more effective overall, the choice of model can significantly impact performance when working with Banglish text.

6. Conclusion and Future Work

Conclusion

This study presents a comparative analysis of four transformer-based models—XLM-RoBERTa, BanglaBERT, BanglaT5, and mT5—for emotion classification in Bangla and Banglish text. Based on the experimental results across multiple evaluation metrics (Accuracy, Precision, Recall, F1 Score, and AUC-ROC), several clear patterns emerged:

- XLM-RoBERTa consistently outperforms all other models in both Bangla and Banglish datasets. It achieves top scores across all metrics and emotion classes, and demonstrates strong generalization even in code-mixed contexts. Its balanced performance across languages, especially high AUC-ROC scores (~0.90+), makes it the most robust and reliable model overall.
- BanglaBERT, while highly effective in native Bangla text due to its language-specific pretraining and morphological sensitivity, struggles significantly with Banglish input. It performs well in AUC-ROC but shows notable drops in core classification metrics for code-mixed data. This highlights its specialization for monolingual Bangla NLP tasks.
- BanglaT5 and mT5 deliver mid-range performance. While not on par with the discriminative models (XLM-RoBERTa and BanglaBERT) in either language, they remain stable across most metrics and can serve as fallback models in multilingual or generative NLP pipelines. However, their performance suffers more in Banglish settings, particularly in emotion classes like *Sadness* and *Surprise*.
- Across all models, the AUC-ROC metric consistently scores the highest, indicating that while classification thresholds vary, the models still retain strong discriminative power.
- Per-class accuracy analysis further validates that XLM-RoBERTa excels in detecting high-intensity emotions such as *Fear*, *Joy*, and *Surprise*, while BanglaBERT offers strong performance in native Bangla but loses effectiveness in code-mixed settings.

Overall, the results confirm that model architecture, training corpus, and language specialization significantly affect performance, especially when handling complex real-world linguistic variations like Banglish.

Future Work

While this research demonstrates the strengths and limitations of current Bangla NLP models, several opportunities exist for further enhancement:

1. **Banglish-Specific Pretraining:** Develop or fine-tune models explicitly on code-mixed Banglish corpora, including social media data, chat logs, and Romanized Bangla content. This could significantly boost performance for models like BanglaBERT in multilingual or informal contexts.
2. **Linguistic Adaptation Techniques:** Implement advanced techniques such as adversarial training, transliteration augmentation, or multi-view learning to improve robustness in low-resource or code-mixed scenarios.
3. **Lightweight Deployment:** While XLM-RoBERTa performs best, it is computationally heavy. Exploring **model distillation**, **quantization**, or training **lighter variants** of BanglaBERT could make these models suitable for real-time or mobile-based NLP applications in Bangladesh.
4. **Emotion Imbalance Handling:** Address class imbalance (common in emotions like *Surprise* or *Disgust*) through data augmentation, weighted loss functions, or oversampling to ensure more equitable performance across all emotion categories.
5. **Multimodal Emotion Detection:** Future research could also explore combining **text with speech, image, or video** for richer emotion detection—particularly valuable in social media, customer support, and education platforms.

By addressing these directions, future work can significantly advance emotion classification in Bangla NLP, making models more inclusive, robust, and ready for deployment in diverse real-world applications.